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JSON UI Documentation

Expand table

JSON UI	Description
ui_defs	The _ui_defs.json file that lists all UI screen files that should be loaded by Minecraft.
ui_element	A UI element definition in a Minecraft Bedrock UI screen file.
ui_global_variables	The _global_variables.json file that defines global variables used across all UI screens in Minecraft.
ui_screen	A UI screen definition file for Minecraft Bedrock Edition.

ⓘ **Note:** The author created this article with assistance from AI. Learn more

Last updated on 01/28/2026

JSON UI Documentation - minecraft:ui_defs

The _ui_defs.json file that lists all UI screen files that should be loaded by Minecraft. This file is required for custom UI screens to be recognized.

UI Definitions Properties

Expand table

Name	Default Value	Type	Description	Example Values
ui_defs	*not set*	Array of strings	An array of paths to UI JSON files that should be loaded. Paths are relative to the resource pack root and typically start with 'ui/'.	Basic: `["ui/my_screen.json","ui/my_common_elements.json"]`

Samples

Basic

JSON

```
{
  "ui_defs": [
    "ui/my_screen.json",
    "ui/my_common_elements.json"
  ]
}
```

ⓘ **Note:** The author created this article with assistance from AI. Learn more

Last updated on 01/28/2026

JSON UI Documentation - minecraft:ui_element

Summarize this article for me

A UI element definition in a Minecraft Bedrock UI screen file. UI elements define the visual and interactive components of Minecraft's user interface.

UI Element Properties

Expand table

Name	Default Value	Type	Description
allow_clipping	*not set*	Boolean true/false	Whether this element can be clipped by parent elements.
alpha	*not set*	Decimal number	The opacity of the element, from 0.0 (transparent) to 1.0 (opaque).
always_handle_pointer	*not set*	Boolean true/false	Whether pointer events are always handled by this element.
anchor_from	*not set*	Anchor From choices	The anchor point on the parent element from which this element is positioned.
anchor_to	*not set*	Anchor To choices	The anchor point on this element used for positioning.
animation_reset_name	*not set*	String	The name of the animation state to reset to.
anims	*not set*	Array of strings	Animation references to apply to this element.
bilinear	*not set*	Boolean true/false	Whether to use bilinear filtering for textures.
bindings	*not set*	Array of Bindings items	Data bindings that connect UI element properties to game data.
button_mappings	*not set*	Array of Button Mappings items	Mappings from input buttons to actions.

Name	Default Value	Type	Description
clip_pixelperfect	*not set*	Boolean true/false	Whether clipping is pixel-perfect.
clips_children	*not set*	Boolean true/false	Whether child elements are clipped to this element's bounds.
collection_name	*not set*	String	The name of the collection to use for grid or stack panel items.
color	*not set*	Array of numbers	The color tint applied to the element, specified as [r, g, b] with values 0.0-1.0.
consume_event	*not set*	Boolean true/false	Whether input events are consumed by this element.
controls	*not set*	Array of Controls items	Child elements contained within this element.
default_focus_precedence	*not set*	Boolean true/false	Priority for receiving default focus.
enabled	*not set*	Boolean true/false	Whether this element is enabled and can be interacted with.
factory	*not set*	String	The factory configuration for dynamic element creation.
fill	*not set*	String from a list of choices	Whether image fills its container.
focus_change_down	*not set*	String from a list of choices	The element to focus when navigating down.
focus_change_left	*not set*	String from a list of choices	The element to focus when navigating left.
focus_change_right	*not set*	String from a list of choices	The element to focus when navigating right.
focus_change_up	*not set*	String from a list of choices	The element to focus when navigating up.
focus_enabled	*not set*	Boolean true/false	Whether this element can receive focus.
font_scale_factor	*not set*	Boolean true/false	A multiplier for the font size.

Name	Default Value	Type	Description
font_size	*not set*	Font Size choices	The size of the font for text rendering.
font_type	*not set*	Font Type choices	The font to use for text rendering.
grayscale	*not set*	Boolean true/false	Whether the element is rendered in grayscale.
grid_dimension_binding	*not set*	String	The binding for determining grid dimensions.
grid_item_template	*not set*	Integer number	The template element to use for grid items.
grid_rescaling_type	*not set*	Grid Rescaling Type choices	How the grid resizes its cells.
ignored	*not set*	Boolean true/false	If true, this element is ignored during layout and rendering.
jump_to_bottom_on_update	*not set*	Boolean true/false	Whether to auto-scroll to bottom when content updates.
keep_ratio	*not set*	Boolean true/false	Whether to maintain aspect ratio when scaling.
layer	*not set*	Integer number	The rendering layer for this element. Higher values render on top of lower values.
locked_control	*not set*	Boolean true/false	The control to display when this element is locked.
max_length	*not set*	Integer number	Maximum character length for text input elements.
max_size	*not set*	Array of numbers	The maximum size constraints for the UI element.
min_size	*not set*	Array of numbers	The minimum size constraints for the UI element.
modal	*not set*	Boolean true/false	Whether this element acts as a modal overlay.
nine_slice_buttom	*not set*	Integer number	The bottom size for nine-slice image rendering.

Name	Default Value	Type	Description
nine_slice_left	*not set*	Integer number	The left size for nine-slice image rendering.
nine_slice_right	*not set*	Integer number	The right size for nine-slice image rendering.
nine_slice_top	*not set*	Integer number	The top size for nine-slice image rendering.
offset	*not set*	Array of numbers	The offset of the UI element from its anchor position. Specified as [x, y] in pixels.
orientation	*not set*	Orientation choices	The orientation for stack panels, determining the direction children are laid out.
place_holder_text	*not set*	String	Placeholder text shown when an input is empty.
prevent_touch_input	*not set*	Boolean true/false	Whether touch input is blocked by this element.
propagate_alpha	*not set*	Boolean true/false	Whether alpha transparency is propagated to child elements.
renderer	*not set*	String	The renderer to use for custom element types. Examples include 'hud_player_renderer', 'inventory_item_renderer', etc.
scroll_box_and_track_panel	*not set*	String	Reference to the panel containing scrollbar box and track.
scroll_content	*not set*	String	Reference to the scrollable content element.
scroll_speed	*not set*	String from a list of choices	The speed of scrolling in scroll views.
scroll_view_port	*not set*	String	Reference to the scroll viewport element.
scrollbar_box	*not set*	String	Reference to the scrollbar box element.
scrollbar_touch_button	*not set*	Boolean true/false	Whether the scrollbar responds to touch.
scrollbar_track	*not set*	String	Reference to the scrollbar track element.

Name	Default Value	Type	Description
shadow	*not set*	Boolean true/false	Whether to render a shadow behind text.
size	*not set*	Array of numbers	The size of the UI element. Can be specified as [width, height] where each can be a number (pixels), percentage string ('100%'), or special value ('fill', '100%c', '100%cm').
slider_range	*not set*	Decimal number	The range for slider elements.
slider_select_on_hover	*not set*	String	Whether the slider value changes on hover.
slider_steps	*not set*	Decimal number	The number of discrete steps in a slider.
text	*not set*	String	The text content for label elements. Can include localization keys.
text_alignment	*not set*	Text Alignment choices	The horizontal alignment of text within a label.
text_box_name	*not set*	String	The name identifier for a text box.
texture	*not set*	String	The texture path for image elements. Relative to the resource pack textures folder.
tiled	*not set*	Boolean true/false	Whether the texture is tiled when rendering.
tts_control_type_order_priority	*not set*	String	The priority for ordering elements in text-to-speech.
tts_ignore_count	*not set*	Boolean true/false	Whether to ignore this element when counting for text-to-speech.
tts_skip_message	*not set*	Boolean true/false	Whether to skip this element in text-to-speech narration.
type	*not set*	Type choices	The type of UI element. Determines how the element is rendered and behaves.
uv	*not set*	Array of numbers	UV coordinates for texture mapping, specified as [u, v].
uv_size	*not set*	Array of numbers	The size of the UV region to sample from the texture.

Name	Default Value	Type	Description
variables	*not set*	Key/item pairs of Variables items	Variable definitions for conditional logic.
virtual_keyboard_buffer_control	*not set*	Boolean true/false	Whether this input uses a virtual keyboard buffer.
visible	*not set*	Boolean true/false	Whether this element is visible by default.

Anchor From choices

Expand table

Value	Title	Description
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		

Bindings

Bindings Properties

JSON path: `bindings`

Expand table

Name	Default Value	Type	Description
binding_collection_name	*not set*	String	The name of the collection when using collection bindings.
binding_condition	*not set*	Binding Condition choices	When the binding should be evaluated.
binding_name	*not set*	String	The name of the data source to bind to (e.g., '#player_name').
binding_name_override	*not set*	String	The property on the UI element to bind the value to.
binding_type	*not set*	Binding Type choices	The type of binding.
source_control_name	*not set*	String	The name of the source control for view bindings.
source_property_name	*not set*	String	The property name on the source control.
target_property_name	*not set*	String	The property name on the target element.

Binding Condition choices

Expand table

Value	Title	Description
undefined		
undefined		
undefined		
undefined		
undefined		

Button Mappings

Button Mappings Properties

JSON path: `button_mappings`

Expand table

Name	Default Value	Type	Description
from_button_id	*not set*	String	The input button that triggers the action.
input_mode_condition	*not set*	Input Mode Condition choices	The input mode required for this mapping.
mapping_type	*not set*	Mapping Type choices	The type of button mapping.
to_button_id	*not set*	String	The action button to trigger.

Input Mode Condition choices

Expand table

Value	Title	Description
undefined		
undefined		

Mapping Type choices

Expand table

Value	Title	Description
undefined		
undefined		
undefined		
undefined		

Controls

Controls Properties

JSON path: `controls`

Expand table

Name	Default Value	Type	Description
[a-zA-Z0-9_.:@]+	*not set*	Object	A child UI element. The key is the element name, optionally with @base_element syntax for inheritance.

Grid Rescaling Type choices

Expand table

Value	Title	Description
undefined		
undefined		
undefined		

Type choices

Expand table

Value	Title	Description
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		

Value	Title	Description
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		
undefined		

Variables

Variables Properties

JSON path: `variables`

Expand table

Name	Default Value	Type	Description
requires	*not set*	Boolean true/false	The condition that must be met for this variable to be active.

ⓘ **Note:** The author created this article with assistance from AI. Learn more

Last updated on 03/05/2026

JSON UI Documentation - minecraft:ui_global_variables

The _global_variables.json file that defines global variables used across all UI screens in Minecraft. Variables are defined with '$' prefix and can include colors, sizes, and other values.

> **Note**
>
> Variable names must start with '$'. Colors are typically specified as [r, g, b] or [r, g, b, a] arrays with values between 0.0 and 1.0.

UI Global Variables Properties

Expand table

Name	Default Value	Type	Description	Example Values
$background_color	*not set*	String		Colors: `[0.2,0.2,0.2,0.8]`
$button_text_color	*not set*	String		Colors: `[1,1,1]`
$highlight_color	*not set*	String		Colors: `[1,0.84,0]`

Samples

Colors

JSON

```
{
  "$button_text_color": [
    1,
    1,
    1
  ],
  "$background_color": [
    0.2,
    0.2,
    0.2,
    0.8
  ],
  "$highlight_color": [
    1,
```

```
    0.84,
    0
  ]
}
```

ⓘ **Note:** The author created this article with assistance from AI. Learn more

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Last updated on 01/28/2026

JSON UI Documentation - minecraft:ui_screen

Summarize this article for me

A UI screen definition file for Minecraft Bedrock Edition. UI screens define complete interface layouts including namespaces, element definitions, and global variables.

> **Note**
>
> UI screen files use a namespace-based structure where each JSON key (except 'namespace') defines a UI element. Elements can inherit from other elements using the '@' syntax (e.g., 'my_button@common.button'). Variables are defined with '$' prefix and can be overridden.

UI Screen Properties

Expand table

Name	Default Value	Type	Description
namespace	*not set*	String	The namespace for this UI screen. Used to reference elements from other UI files using the 'namespace.element_name' syntax.

Note: The author created this article with assistance from AI. Learn more

Last updated on 03/05/2026